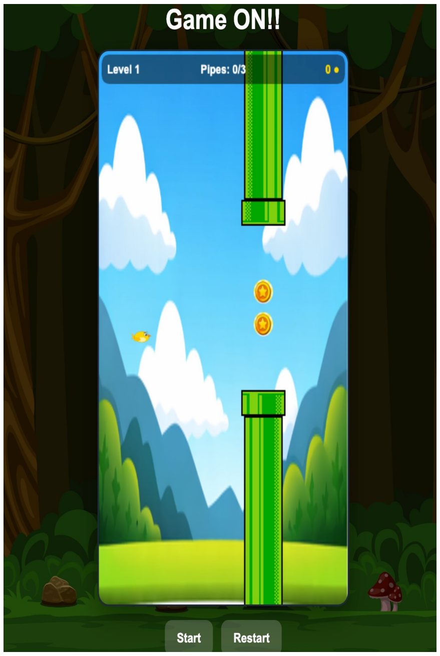
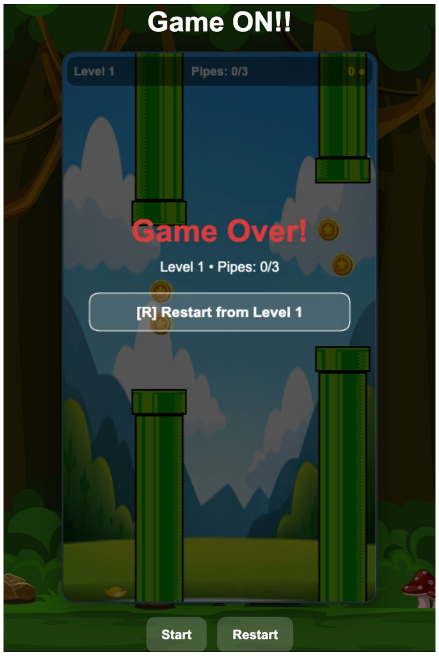
A browser-based arcade game built with Vue 3 + Canvas API



Gameplay



Game Over

Victory Screen

■ Project Overview

About the Game

Game ON!! is a polished, feature-rich Flying Bird Game built from scratch using Vue 3 and the HTML5 Canvas API. The game features a complete 5-level progression system, collectible coins, dynamic sound, smooth animations, and celebratory win screens, all delivered as a single Vue component with zero external game libraries.

Tech Stack

Framework	Vue 3 (Composition API + script setup)
Rendering	HTML5 Canvas 2D Context
Animation	requestAnimationFrame game loop with delta-time
Audio	Web Audio API — coin, game-over, win, background music
Assets	Sprite images (bird, pipes, coins, background)
Input	Keyboard (Space / R / C) + Pointer (tap/click)
State	Single reactive game object + Vue ref for UI sync

Key Metrics

Lines of Code	926 lines (single .vue component)
Game Levels	5 progressively harder levels
Canvas Size	360 × 640 px (mobile-optimised portrait)
Audio Tracks	4 sounds: coin collect, game over, win, background music

■ Core Features

5-Level Progression	Each level ramps up speed, narrows the pipe gap, and increases the number of pipe pairs to survive. Level 1 is gentle (gap 190 px, speed 2.2); Level 5 is intense (gap 130 px, speed 3.4, 7 pipe pairs).
Coin Collection System	Coins spawn in the gap between pipes. Players collect them mid-flight. Minimum coins per level range from 4 (L1) to 12 (L5). Coins carry over across levels and are shown on the final win screen.
Continue / Lives System	On Game Over from Level 2 onwards, players can spend accumulated coins to continue from the current level. Continue costs escalate: 3 / 5 / 7 / 9 coins.
Hover Animation	When idle (READY state) the bird hovers smoothly using a sine-wave offset, giving the game a polished, living feel before play begins.
Confetti Win Screen	Clearing all 5 levels triggers an animated confetti explosion rendered on canvas, plus a gold trophy card showing total coins collected.
Sound & Music	Background music loops during gameplay. Discrete sound effects fire on coin collection, game over, and victory. Audio is preloaded for zero-latency playback.
Keyboard + Touch Input	Spacebar to flap, R to restart, C to continue — plus full pointer/tap support making the game fully playable on mobile browsers.
HUD Display	A persistent heads-up display shows current level, pipes passed vs required, and coin count — all updated in real time each game frame.

■■ Level Design & Architecture

Level Progression Table

Level	Pipe Gap	Speed (px/frame)	Pipe Pairs	Spawn Delay	Min Coins
1 Beginner	190 px	2.2	3	3200 ms	4
2 Easy	175 px	2.5	4	3000 ms	6
3 Medium	160 px	2.8	5	2800 ms	8
4 Hard	145 px	3.1	6	2600 ms	10
5 Expert	130 px	3.4	7	2400 ms	12

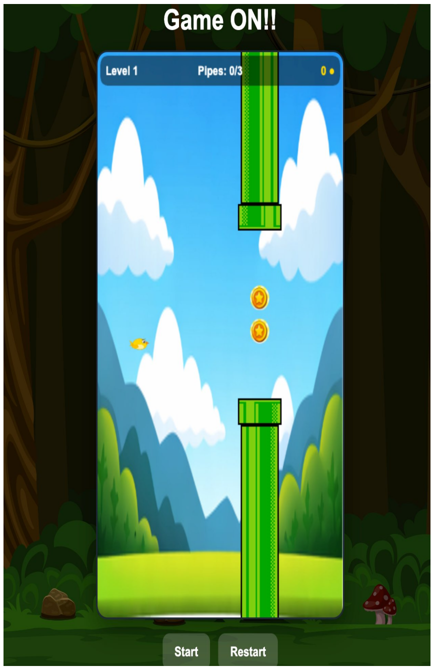
Architecture: Game Loop

The game runs a delta time driven requestAnimationFrame loop. Each tick it: (1) advances physics (gravity, velocity, position), (2) spawns pipes on a millisecond accumulator, (3) moves pipes and coins left, (4) tests AABB collision with pipes and coin overlap, (5) clears the canvas and redraws all sprites, (6) draws the HUD, and (7) handles state transitions (LEVEL_COMPLETE, GAME_OVER, WINNER).

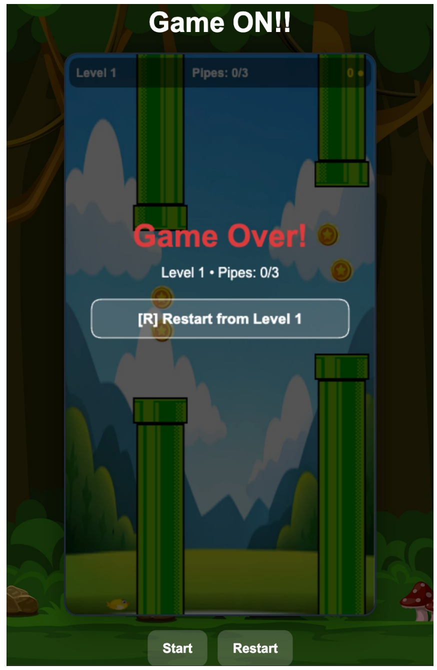
Physics Constants

Gravity	0.35 px/frame²
Flap Velocity	6.0 px/frame upward
Terminal Velocity	8.0 px/frame downward cap
Hover Amplitude	7.0 px sinusoidal (idle only)
Bird Size	34 × 24 px sprite
Pipe Width	64 px

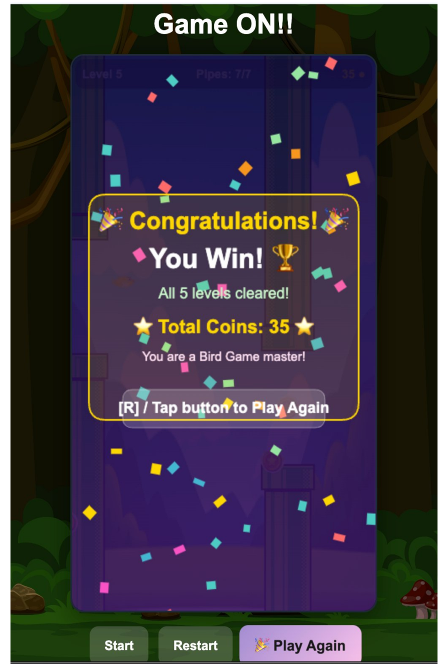
■ Screenshots & Game States



PLAYING State
Bird navigates pipe gaps. HUD shows Level, Pipes passed, and Coin count. Coins appear in the gap for mid-flight collection.



GAME OVER State
Triggered by pipe or floor/ceiling collision. Shows level reached and pipes cleared. From L2+ players may spend coins to continue.



WINNER State
All 5 levels cleared! Confetti rains across the canvas. Gold trophy card shows total coins earned across the run.

State Machine

The game cycles through six states: READY > PLAYING > LEVEL_COMPLETE > PLAYING (next level) > WINNER or GAME_OVER > optional PLAYING via continue. State is stored in a plain JS object (game.state) and mirrored to a Vue ref for reactive template bindings.

■ Summary & Future Roadmap

What Was Achieved

- ■ Complete 5-level playable game in a single Vue component
- ■ Delta time physics ensuring consistent gameplay across frame rates
- ■ Coin economy with cross level persistence and continue mechanic
- ■ 4 audio assets with preloading and looping background music
- ■ Animated confetti particle system on victory
- ■ Mobile friendly pointer input alongside full keyboard controls
- ■ Clean separation of game state and Vue reactivity

Potential Future Enhancements

- ■ Persistent high score leaderboard (localStorage or backend)
- ■ Additional power ups: shield, slow motion, magnet coins
- ■ Bird skin and character selection menu
- ■ Mobile app packaging via Capacitor or PWA manifest
- ■ Accessibility improvements (ARIA, reduced motion mode)
- ■ Multiplayer race mode (WebSocket)
- ■ Level editor for custom pipe layouts

Live Demo

Play the game online at: https://ammun99.github.io/bird-game-page/